

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
8	(a)		$E = \frac{hc}{1.06 \times 10^{-6}} = 1.88 \times 10^{-19} \text{ [J]} \text{ (1)}$ $\div e = 1.18 \text{ [eV]} \text{ (1)}$ $1.43 - 1.18 = 0.25 \text{ [eV]} \text{ (1)}$		3		3	3	
	(b)		Level P is not metastable / electrons drop to level U which is metastable (1) Population inversion is {not between P and ground / between U and L} (1)		2		2		
	(c)		Any two × (1) from: • More efficient / require less energy / pumped electrically with lower voltage • Cheaper • Mass produced • Smaller / more compact Don't accept safer / more environmentally friendly / don't overheat	2			2		

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	(d)		Any two × (1) from: • Toxic chemicals/Pb/Hg/Cd/Ba could get into ground water / contaminate soil / crops / livestock / wildlife • Toxic chemical / Br could be released into air (when warmed) • Landfill sites decreases value of surrounding land / lack of land for housebuilding • Landfill sites nearing capacity • Landfill sites create visual pollution • Replacement electrical devices contributing to CO₂ emissions • Provide opportunity for [recycling] jobs • Scarce resources e.g. tantalum wasted / rare metals not being reused • Wasted energy extracting new resources e.g. mining Accept any other valid points			2	2		
			Question 8 total	2	5	2	9	3	0